

Hospital Patient & Billing Analytics Dashboard

Detailed Functional Specifications, Dashboard Architecture & Logic

1. PROJECT TITLE

Hospital Patient & Billing Analytics Dashboard

2. DASHBOARD SECTIONS

- **Overview / KPI Summary:** headline metrics at a glance
- **Patient Demographics:** age, gender, blood type breakdown
- **Admissions Analysis:** trends by condition, admission type, over time
- **Billing & Insurance:** revenue trends, billing by insurance provider
- **Clinical Outcomes:** test results distribution, medication usage

3. DATA REQUIREMENTS

Section	Fields Needed
KPI Summary	Billing Amount, Length of Stay, Name (count)
Demographics	Age, Gender, Blood Type
Admissions	Date of Admission, Medical Condition, Admission Type, Hospital
Billing & Insurance	Billing Amount, Insurance Provider, Date of Admission
Clinical Outcomes	Test Results, Medication, Medical Condition

4. FILTERS / SLICERS

- Date range (Date of Admission)
- Medical Condition
- Admission Type (Emergency, Elective, Urgent)
- Insurance Provider
- Gender

5. VISUALS / CHARTS

- KPI Cards: Total Patients, Total Billing, Avg Length of Stay, Avg Billing per Patient
- Bar chart: Admissions by Medical Condition
- Line chart: Admissions trend over time (monthly)

- Donut chart: Admission Type distribution
- Stacked bar: Billing by Insurance Provider
- Bar chart: Test Results distribution
- Table: Top hospitals by patient volume

6. INTERACTIVITY

- Cross-filtering between all visuals on click
- Drill-down from Year → Month → Day on the admissions trend chart
- Slicers apply globally across all dashboard pages

7. CALCULATIONS / MEASURES

Measure Name	Formula / Expression
Total Billing Amount	<code>SUM(Billing Amount)</code>
Average Billing per Patient	<code>AVERAGE(Billing Amount)</code>
Total Patients	<code>DISTINCTCOUNT(Name)</code>
Average Length of Stay	<code>AVERAGE(Length of Stay)</code>
% Abnormal Test Results	<code>COUNT(Test Results="Abnormal") / COUNT(Test Results)</code>

8. EXPORT / REPORTING NEEDS

- Export to PDF for stakeholder distribution
- Export underlying tables to Excel for finance team review

9. NOTES / SPECIAL INSTRUCTIONS

- Dashboard should support both executive summary view and drill-down detail view
- Color-code Admission Type consistently across all visuals (Emergency=red, Urgent=orange, Elective=green)